

Calculus I – Syllabus

Learning Targets: The learning targets are organized by topic and listed below.

Foundation

1. Find and interpret average velocity and its units using information about the position function (as a table, graph, and/or function formula).
2. Find the limit of a function at a point using numerical, graphical, and algebraic methods.
3. Find the derivative of a function using the limit definition of the derivative.
4. Use derivative notation correctly, state the units of a derivative, and correctly interpret the meaning of a derivative in context.
5. Find the equation of the tangent line to a function at a point and use the tangent line to estimate values of the function.
6. Determine where a function is continuous or differentiable given a graph or formula of the function and explain the reasoning.
7. Sketch a function or its derivative given information about either the original function, the first derivative, or the second derivative.

Derivatives Rules

8. Compute derivatives correctly for power, polynomial, and exponential functions and the sine and cosine functions, and basic combinations of these (constant multiples, sums, differences).
9. Compute derivatives correctly for products, quotients, and composites of functions.
10. Compute the derivatives correctly for logarithmic, trigonometric, and inverse trigonometric functions.
11. Determine which derivative rules to use and compute derivatives correctly using multiple rules in combination
12. Compute the derivative of an implicitly-defined function and find the slope of the tangent line to an implicit curve.

Applications

13. Set up and use derivatives to solve related rates problems.
14. Identify limits in indeterminate form and apply L'Hospital's Rule to evaluate them.
15. Use the Extreme Value Theorem to find the absolute maximum and minimum values of a continuous function on a closed interval.
16. Find the critical values of a function, determine where the function is increasing and decreasing, and apply the First Derivative Test to classify the critical points as local extrema.
17. Determine the intervals of concavity of a function and find all of its points of inflection.
18. Set up and use derivatives to solve applied optimization problems.

Integrals

19. Find the antiderivative of basic functions.
20. Estimate areas and integrals using Riemann sums.
21. Explain the meaning of each part of the definition of the definite integral in terms of a graph, and interpret the definite integral in terms of areas, net change, and displacement.
22. Evaluate a definite integral using geometric formulas and the Properties of the Definite Integral.
23. Evaluate a definite integral using the Fundamental Theorem of Calculus
24. Find the net change of a function over an interval using a definite integral.

Other

25. Score at least 70% on every homework assignment or earn 80% of all homework points.
26. Satisfactorily complete at least 70% of labs.

Grading: Your grade in this class will be computed using a mastery-based grading scheme. This type of grading scheme has also been called target-based grading, standards-based grading, or competency-based grading. Your score will be the number of learning targets you master, plus your score on the final exam for a maximum of 36 points. You need to earn at least 26 points to pass the class. The following chart outlines how many points you need to earn each grade.

Letter	A+	A	A-	B+	B	B-	C+	C	C-	D+	D
Points	36	35	34	33	32	31	30	29	28	27	26

Although the final exam will have 15 questions, the maximum you can earn is 10. For example, if you complete 20 learning targets and successfully answer 12 questions on the final exam, you will earn 30 points and receive a C+.

Meeting a Learning Target: There are 26 learning targets included earlier. You need to reach *Met* twice to earn credit for each of the first 24 learning targets. The homework and lab learning targets do not need to be completed twice.

Learning Target Grading Rubric: Each attempt to master one of the first 24 learning targets will be graded on the following rubric.

- Met:* Mathematics is correct, and the write-up is clear, detailed, and easy to follow with proper notation and terminology.
- Revise:* Your explanation/mathematics needs some minor revisions. Submit a revised solution to get to Met. Revisions must be submitted by the next class period after the work was returned.
- Unmet:* There is at least one serious flaw in your mathematics. When a new assessment is given, try again to reach Met.

Teamwork: During class time, you will regularly work on problems in teams. This work will be collected to count towards mastery for a specific learning target.

Labs: Some class time will be devoted to labs. You will spend the class working on a lab with a team to be collected at the end of the session. Completing 70% of these labs satisfactorily is one of your learning targets.

Assessments: Approximately every other week, we will have an in-class assessment. You will work individually to solve learning targets.

Final Exam: The final exam will cover questions from 15 learning targets. Two questions may be graded as *Revise*, but the rest must be *Met*. For example, if you answer 7 questions at Met and 4 at Revise, this will count as a 9 on the final. Meeting expectations on a learning target during the final exam will also count towards mastery of that learning target. Students who need and have earned extra attempts on any of the other 9 learning targets will also be provided questions on those topics. The maximum score you can earn on the final exam is 10.

Calculus I – Sample Instruction

I planted a tomato plant in my backyard.

h = the height in inches of the plant at season's end.

Question: What are some variables that might impact how tall the plant grows?

t =

v = the gallons of water the plant gets

p =

Question We want to see how the height of the plant changes based on the gallons of water the plant gets.

a. What should we compute?

b. What are its units?

c. Would you expect it to be positive or negative? Give reasoning.

Calculus I – Sample Instruction

Learning Target - Derivatives 11: Compute the derivatives correctly for logarithmic, trigonometric, and inverse trigonometric functions.

Question: Earlier you computed the derivative of $f(x) = \frac{\sin x}{\cos x}$ using the quotient rule. You found that $f'(x) = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$.

a. What function is $f(x) = \frac{\sin x}{\cos x}$?

b. Let's simplify $f'(x)$.

Derivative of all Trig Functions:

$$\frac{d}{dx}(\sin x) =$$

$$\frac{d}{dx}(\cos x) =$$

$$\frac{d}{dx}(\sec x) =$$

$$\frac{d}{dx}(\csc x) =$$

$$\frac{d}{dx}(\tan x) =$$

$$\frac{d}{dx}(\cot x) =$$

Derivative of all Inverse Trig Functions:

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\csc^{-1} x) = \frac{-1}{|x|\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

Question: On the previous page we had the derivative of all the trig functions and all the inverse trig functions. You should have all the trig derivatives memorized. However, I only expect you to have one inverse trig derivative memorized. That one, in particular, will be useful in Calculus II. Which inverse trig derivative do you think it is?

Question: What's the difference between $\sin^{-1} x$ and $\arcsin x$?

Derivative of natural log:

$$\frac{d}{dx}(\ln x) =$$

All logarithm functions:

$$\frac{d}{dx}(\log_b x) =$$

Calculus I – Sample Teamwork

Scribe: _____

Herald: _____

Guardian: _____

Navitagor: _____

1. Compute the derivative of $f(x) = \frac{e^x}{\tan(x)}$.

2. Compute the derivative of $f(x) = \sec x$.

3. Compute the second derivative of $\sec x$.

4. Compute the derivative of $y = \log_5 x$.

5. Compute $\frac{d}{dt} (\ln(3t^2 - 5))$.

Note: You will need to use the chain rule.

6. Compute $\frac{dg}{d\theta}$ where $g(\theta) = \arctan \theta \ln \theta$.

7. Compute the derivative of $f(x) = x \sin^{-1} x$

8. A \$1000 deposit earns 5% interest compounded annually. The time to reach a balance of A dollars is $t(A) = \log_{1.05}(A/1000)$.
- Calculate $t'(A)$. (Note, you will need to use the chain rule.)
 - Calculate $t'(2000)$. Write a complete sentence with correct units that explain what this value represents.
 - Calculate $t'(2100)$. Write a complete sentence with correct units that explains what this value represents.

9. You need to build a ramp from the ground to a porch that is 3ft above ground level. The length of the ramp is a function of the angle of rise, θ . The length of the ramp is given by $L(\theta) = 3\csc(\theta)$.

(a) Calculate $L'(\theta)$.

(b) Calculate $L'(30)$. (Put your calculators in degrees.)

(c) In one sentence, explain what your answer in part b tells us.

10. Investing x thousand dollars in advertising generates revenue $R(x) = 50\ln(x + 1)$.
- Calculate $R'(x)$
 - How is the revenue changing when we spend 30 thousand dollars on advertising?
Your answer should include a number with units.
 - How is the revenue changing when we spend 55 thousand dollars on advertising?
Your answer should include a number with units.
 - Compare your answers to parts b and c. What does this tell us about the rate of change in revenue at the two points?

Calculus I – Sample Teamwork

Scribe: _____

Herald: _____

Guardian: _____

Navitagor: _____

1. Suppose we wanted to know how the height of the plant, h , changes based on the number of ounce my dog pees on it, p .

- a. What would we compute?

- b. What are its units?

- c. Would you expect it to be positive or negative? Give reasoning.

Calculus I – Sample Lab

MTH 141 Week 6 - Motion Detector

Part One

Scribe: _____

Herald: _____

Guardian: _____

Navigator: _____

Directions:

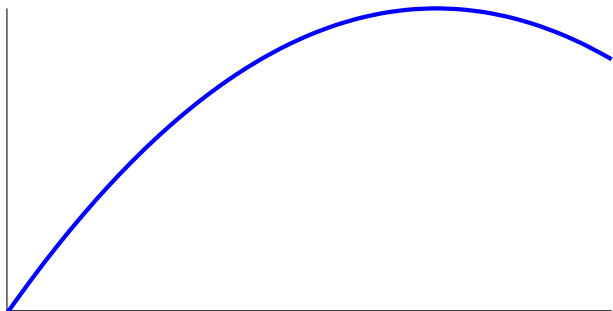
1. Set up the track and cart.

Practice:

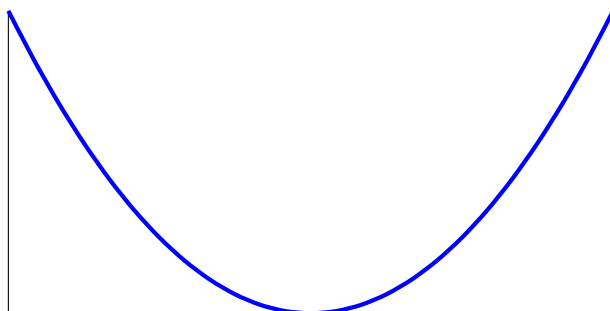
1. With your sensor on the track, click the 'collect' button near the top of your screen and move the cart slowly back and forth on the track either until the collection stop stops or you choose to manually stop by hitting the 'stop' button at the top of the screen.
2. To begin a new graph you can either 'hit' the collect button or choose 'start a new experiment' using the top left button.
3. Each group member should take a turn creating a graph by moving the cart on the track.
4. If you are only collecting the position, look at the tab to the left of the graph. If you click there, you can have the app collect both the position and the velocity graph. Change the setting so that you collect the position function in blue and the velocity graph in red. Now begin a new graph collecting both the position and the velocity.

Part One: Below you see eight position graphs. Move the cart to make a graph that matches each of ones below. Once you have created the graph, have either Violet or Dr. Lane initial your graph. The graphs must be completed in order. If your team has three members, rotate the role of navigator.

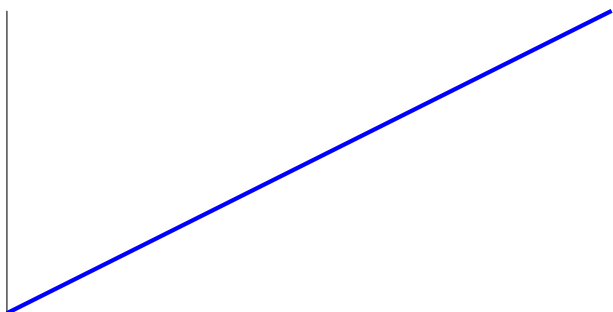
Graph 1 - Scribe



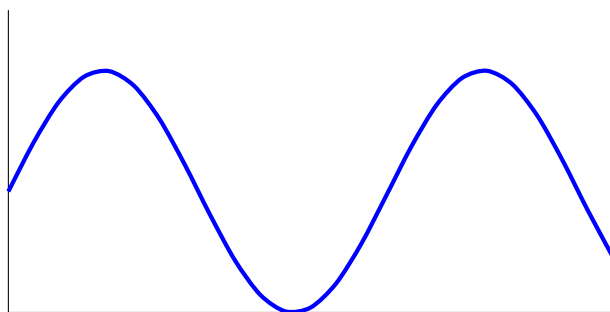
Graph 2 - Herald



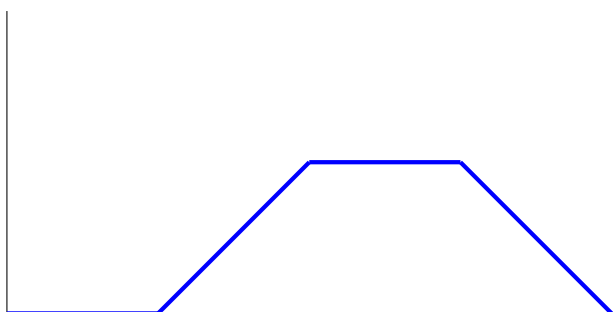
Graph 3 - Guardian



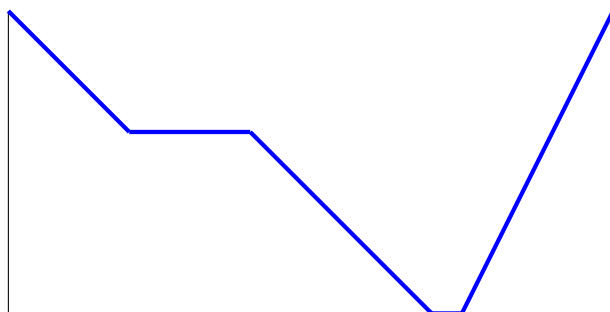
Graph 4 - Navigator



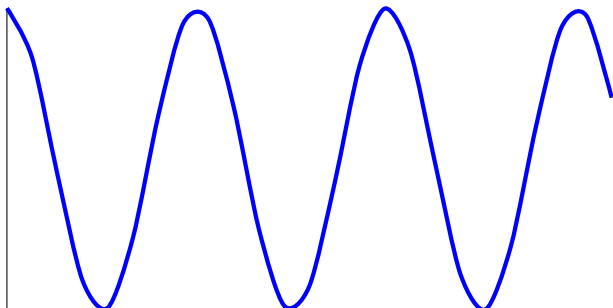
Graph 5 - Scribe



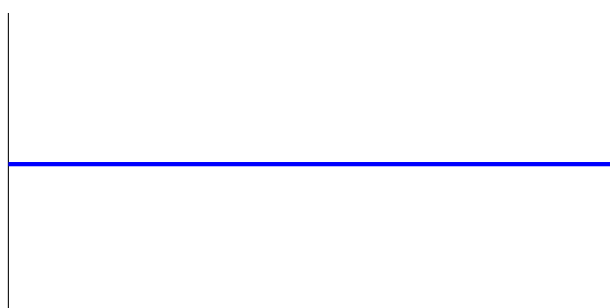
Graph 6 - Herald



Graph 7 - Guardian



Graph 8 - Navigator

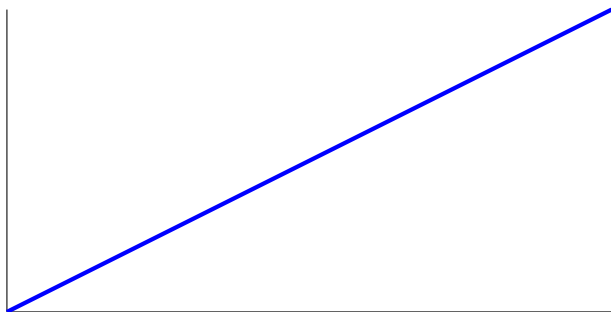


Answer the following question and ask either Violet or Dr. Lane to approve your responses before you move to the next page.

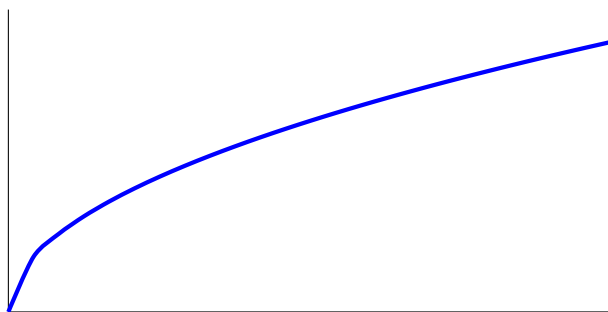
1. Why could we not create any graphs in which the position function went below the x-axis?
2. Describe how you moved the cart to get a slope of zero on the position graph.
3. Describe how you moved the cart to get a positive slope on the position graph.
4. Describe how you moved the cart to get a negative slope on the position graph.
5. Describe how you moved the cart a constant slope on the position graph.

Part Two: Below you see eight velocity graphs. Move the cart to make the graph that matches each of the ones below. When saving your graphs, make sure you only have the velocity function displayed. Once you have created the graph have either Dr. Strunk or Dr. Lane initial your graph. The graphs must be completed in order. If your team has three members, rotate the role of navigator.

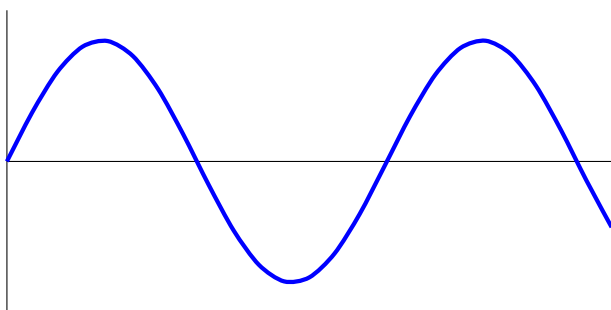
Graph 9 - Scribe



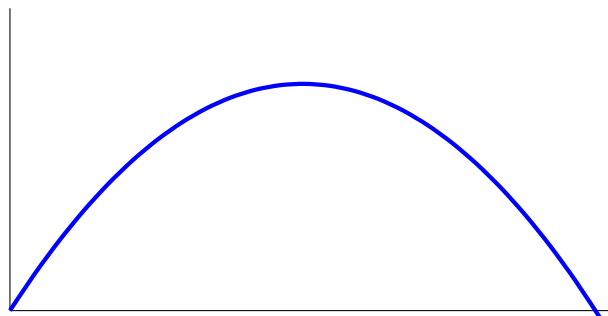
Graph 10 - Herald



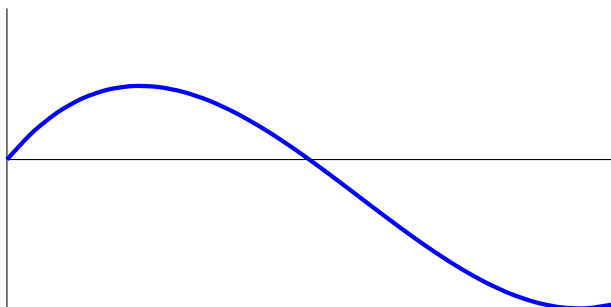
Graph 11 - Guardian



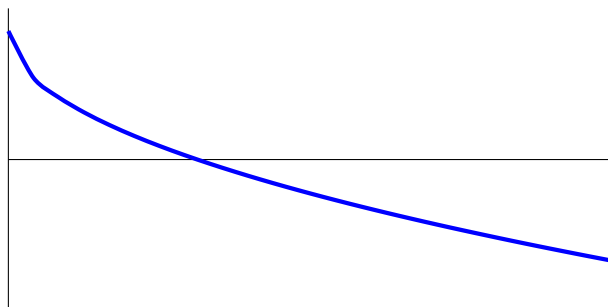
Graph 12 - Navigator



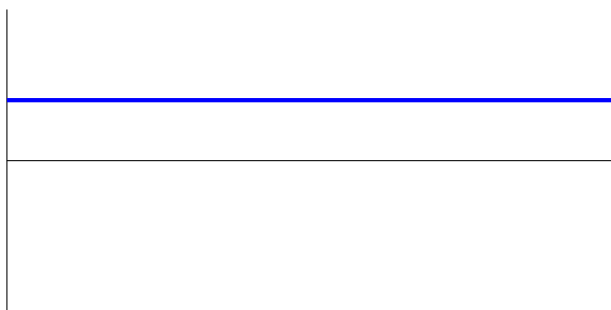
Graph 13 - Scribe



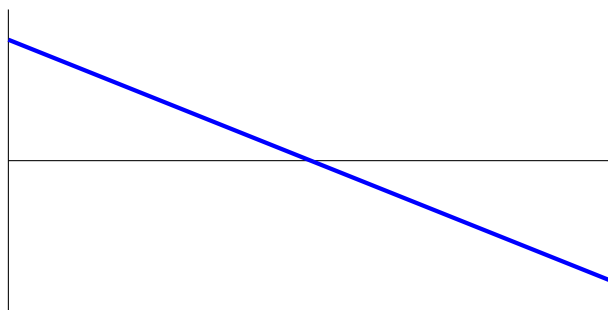
Graph 14 - Herald



Graph 15 - Guardian



Graph 16 - Navigator



Answer the following question and ask either Violet or Dr. Lane to approve your responses before you move to the next page.

1. Unlike our position graphs, we can create velocity graphs where the graphs goes below the x-axis. Why could we do it for the velocity graphs, but not the position graphs?
2. Describe how you would move the cart to get a zero value on the velocity graph.
3. Describe how you moved the cart to get a negative value on the velocity graph.
4. Describe how you moved the cart to get a positive value on the velocity graph.
5. Describe how you moved the cart to get a positive slope on the velocity graph.
6. Describe how you moved the cart to get a negative slope on the velocity graph.
7. Describe how you moved the cart a constant slope on the velocity graph.

Calculus I – Sample Lab

MTH 141 Week 6 - Newton's Method

Scribe: _____

Herald: _____

Guardian: _____

Navigator: _____

After learning about quadratic functions, one of the first things you learned was how to find its roots (or zeros). That is, you learned how to find that x -intercepts.

Example: Find the x -intercepts of the function $f(x) = x^3 - x^2$ using algebra. Verify by looking at a graph of the function.

Background As long as our function is *nice*, we can use basic algebra to find the zeros, but what if it isn't nice? Sir Issac Newton devised a method for approximating the roots of functions. Newton would estimate where a zero might be and then calculate the tangent line to the graph at that point. He would then use the x -intercept of the tangent line to be an improved estimation of the actual zero. He would repeat this process as many times as necessary.

Newton's Method:

Step 1: Take a guess for where the root might be, call this x_0 .

Step 2: Find the tangent line at that spot.

Step 3: Find the x -intercept of the tangent line. Call it x_1 .

Step 4: The answer to the previous question, x_1 , is your new guess for the root, go back to Step 2. Add 1 to the subscript for each subsequent guess: x_2 , x_3 , etc.

Task One - Exploring Newton's Method: Let's try Newton's Method with the function we defined above and try to approximate the root at $x = 1$.

Step 1: We start by guessing what the root might be. What is its error? That is, what is the distance between our guess and what we know the answer to be ($x = 1$). You can find this by computing $|1 - x_0|$.

Let's guess $x_0 = 2$. *This feels a little silly, because we already know the root is 1.*

The error is _____

Step 2: Find the tangent line to $f(x)$ at $x_0 = 2$.

Step 3: Find the root of the tangent line. This is our new estimate. (Give this new estimate the label x_1) What is the error of our estimate, i.e what is $|1 - x_1|$?

Step 4: Repeat the process above with x_1 to get a new estimate x_2 . What is the error of this new estimate?

Summary: Summarize your results below, by filling in the table:

i	0	1	2
x_i			
Error			

Desmos: Graph $f(x)$ and the tangent line you found for $x_0 = 2$. Confirm that it is tangent and the x -intercept is the value you found for x_1 .

Desmos: Graph $f(x)$ and the tangent line you found for x_1 . Confirm that it is tangent and the x -intercept is the value you found for x_2 .

Task Two - Defining Newton Recursively Newton's Method can be described by the recursive sequence $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

1. Start with $x_0 = 2$. Use the recursive formula above to find x_1 . Confirm that it matches the previous value you found for x_1 .
2. Use the recursive formula to calculate x_2 . Confirm that it matches the previous value you found for x_2 .
3. Use the formula to find x_3 . What is the error?

Question: Let's try Newton's Method with another function. Consider $f(x) = x^2 - \sin x$.

1. Look at a graph of $f(x)$ and estimate the roots.
2. The obvious root is $x = 0$. Use Newton's Method to estimate the other root. We need to be careful about first estimate x_0 . Why would $x_0 = 0.25$ not be a good first estimate?
3. As a group, choose a value for x_0 and then use Newton's recursive method to find x_1 , x_2 , and x_3 .

i	0	1	2	3
x_i				
Error				

Task Three - Does it always work? Consider the piecewise defined function:

$$f(x) = \begin{cases} \sqrt{x} & \text{if } x \geq 0 \\ -\sqrt{-x} & \text{if } x < 0 \end{cases}$$

1. Graph $f(x)$ in DESMOS and identify the zero.

2. Determine the derivative and fill in the blanks on the chart below.

$$f'(x) = \begin{cases} \underline{\hspace{2cm}} & \text{if } x > 0 \\ \underline{\hspace{2cm}} & \text{if } x < 0 \end{cases}$$

3. Each member of the group should choose a different initial guess x_0 . Make sure a few are greater than 0 and a few are less than 0. Use Newton's Method to calculate the first three approximations of the zero and record your work in the chart.

Member	x_0	x_1	x_2	x_3
Scribe				
Herald				
Guardian				
Navigator				

4. What do you notice?

5. Let's explore what about the previous function is causing Newton's Method to not work.
- Assume that your initial guess $x_0 > 0$. Write an expression for x_1 in terms of x_0 and simplify.
 - Assume that your initial guess $x_0 < 0$. Write an expression for x_1 in terms of x_0 and simplify.
 - What do you notice?
 - How does this help explain why we never got closer to the zero using Newton's Method for this function?
 - Graph the function in Desmos, what do you notice about the derivative at the zero?

Task Four - Create Your Own Challenge: Try to find another function $f(x)$ that Newton's Method fails to approximate the zero. Use the idea that you just discovered in Task Three. It is okay if you are not successful, but show what you tried.